FinPanel: Evidence-Preserving Infrastructure for Reproducible Point-in-Time SEC Fundamentals

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Abstract

Financial statements describe economic periods, but researchers encounter their contents through filings, revisions, and subsequently retrieved data products. Reproducible fundamentals therefore require more than assigning an accounting number to a fiscal label. This paper examines FinPanel as a bounded software artifact for preserving the evidence behind point-in-time SEC fundamental-data decisions. Its architecture separates filing-time eligibility from snapshot retrieval vintage, filters evidence before fiscal interpretation, applies explicit revision policies, and permits quarter reconstruction only under conservative compatibility rules. Resolved values retain provenance; ambiguity, conflict, and insufficient evidence remain explicit results. Immutable snapshots and software-bound receipts support semantic replay. A repository audit recounts an archived FinPanel 0.1.0a1 purposive study of 100 operating companies, six metrics, and 24,000 requested cells. Of these, 16,706 resolve, comprising 14,136 reported and 2,570 derived values: 69.61% scalar coverage, not an accuracy estimate. Historical calendar and revision cases illustrate a shortened H&R Block transition, week-based years, and Mattel operands that either support derivation or trigger refusal under the implemented quarterly contract. Original-source verification succeeds for 13 of 200 separate capture-vintage queries, rather than measuring verification of the 24,000 historical cells. The evidence supports an inspectable implementation of conservative contracts within the audited population. It does not establish historical API delivery, market-wide accounting correctness, representative sampling, or improved investment performance.

Keywords: SEC EDGAR; point-in-time data; temporal eligibility; fiscal calendars; revision policy; provenance; computational reproducibility.

1. Introduction

Revenue may concern December, appear in a February filing, be revised later, and enter a downloaded dataset years afterward. Confusing these dates can make information appear historically available before the evidence supports it. A reproducible pipeline must explain both the selected number and its temporal eligibility.

Reconstructing fourth-quarter net income requires a compatible annual interval, nine-month amount, eligible versions, and coherent operands. Successful subtraction alone establishes neither a reported quarter nor historical availability. An unresolved scalar can be the appropriate outcome.

FinPanel treats each panel cell as an evidence-bearing decision: reported, conservatively derived, or explicitly unresolved. Sources and reasons accompany the outcome. Preserving these decisions lets researchers inspect and repeat dataset construction; it does not make every conservative refusal correct.

The study addresses three research questions. **RQ1:** How can a fundamentals pipeline separate historical filing eligibility, fiscal-period interpretation, revision selection, and retrieval vintage without allowing excluded future evidence to influence admitted decisions? **RQ2:** What does

conservative resolution produce in the repository's bounded operating-company population, including difficult calendar and revision cases? **RQ3:** What can immutable evidence and semantic receipts reproduce, and which claims remain outside that guarantee?

The contribution is an inspectable implementation, explicit temporal and arithmetic contracts, and a critical audit. Prior work addresses look-ahead bias, temporal databases, provenance, and reproducibility; a close working paper also discusses point-in-time accounting pipelines. No first-ever, investment-performance, or comparative-superiority claim is made.

The bounded artifact is FinPanel 0.1.0a2 at Git commit `28dcecce208a578b4a7b507bbf6d93018b4ab1d6`. The archived benchmark instead records 0.1.0a1 and its own software fingerprint. Inspection and regression of the current commit are distinguished throughout from recounting that earlier experiment.

2. Related work and positioning

Banz and Breen's publisher abstract reports that accounting-database selection and look-ahead effects can change portfolio findings. This motivates attention to data construction; it supplies no distortion estimate for FinPanel, which does not repeat their portfolio experiment. [1]

Jensen and colleagues distinguish valid time from transaction time. FinPanel separately treats the economic interval, filing-availability proxy, and retrieval event: downloading an old filing today does not recover upstream services' contemporaneous histories. [2]

SEC documentation distinguishes CompanyFacts aggregates from calendar-aligned Frames and describes delays between dissemination and API updates. Filing eligibility therefore cannot establish historical aggregate delivery. XBRL additionally makes entity, period, and units essential to fact interpretation. These sources motivate context preservation, not certification of FinPanel's implementation. [3; 4]

W3C PROV describes entities, activities, and agents producing data; FAIR emphasizes metadata and reuse. FinPanel's hashes, pointers, and transformation records address related concerns without establishing PROV or FAIR compliance. Local reproducibility can coexist with inaccessible evidence. [5; 6]

The National Academies distinguish computational reproducibility using the same inputs and methods from replication with independently obtained evidence. Semantic replay uses the former meaning: it repeats represented decisions and evidence identity, potentially including errors. It does not establish independent agreement. [7]

Wang's 2026 SSRN abstract describes temporal layers, preserved revisions, quarter-parent lineage, and experiment manifests, substantially overlapping this paper. Full text was not obtained; detailed methods and results remain unexamined. The suggested citation and a first-person public announcement name Xindi Wang, while the linked SSRN profile displays Arthur Wang. That discrepancy remains unexplained. The limited access precludes priority or feature-level distinctions; unexamined methods are not presumed absent. [8; 9]

The contribution is consequently a bounded implementation and validation study. Comparative usefulness requires additional experiments.

3. Temporal model and scope of the guarantee

Let an observation carry an issuer identity, concept, value, unit, represented interval or instant, filing accession, and source provenance. Let a request specify a financial cutoff (t), revision policy (p), source policy (s), and evidence snapshot (S). Its result can be written conceptually as

$$D = F(S, t, p, s, q, c),$$

where (q) contains the requested entity, metric, and period, and (c) denotes software and interpretation contracts. This expression is an explanatory abstraction, not an assertion of formal verification. The result (D) includes a state, any scalar, and the evidence supporting the decision. Changing the snapshot, policy, or contract can change the dataset even when the visible query label stays the same.

The first temporal dimension is the period represented by the fact. Balance-sheet measures describe an instant; income and cash-flow measures describe durations. Neither a period end nor a fiscal-year label establishes when an investor could have known the amount. The second dimension is filing-linked eligibility. FinPanel admits an aware acceptance timestamp at or before the request cutoff. Acceptance is explicitly treated as a proxy, not proof of dissemination or execution opportunity. Where only a filing date is available, the date must precede the cutoff's SEC-local calendar day. Same-day date-only evidence remains excluded. Unknown availability and invalid filing links cannot establish eligibility.

Retrieval vintage identifies the bytes, sources, and collection metadata used. A 2026 snapshot may contain observations linked to older filings. Filtering them by eligibility creates a documented historical view of that capture, not a reconstruction of what CompanyFacts actually served at the historical cutoff. This materially limits the phrase “point-in-time.”

FinPanel removes ineligible evidence before fiscal interpretation and revision selection. Current issuer metadata without historical availability is withheld. Otherwise, a later fiscal-year-end setting could alter earlier classification despite exclusion of later values: interpretive metadata can also leak future information.

first_reported selects earliest eligible candidates within preserved evidence; latest_available selects latest candidates by the cutoff, not today; all_available retains eligible versions. Ties or incomplete ordering can produce conflict. Accessions do not invent temporal order, and missing earlier observations or earnings announcements cannot be inferred.

Original-XBRL verification is separate from aggregate selection. Its contract requires source and discovery retrieval evidence by the verification cutoff. Later inspection cannot be backdated. Accordingly, capture-vintage verification queries are reported separately from historical panel cells.

4. Architecture and decision contracts

4.1 Evidence acquisition and bounded interpretation

The pipeline connects SEC capture, filing timelines, fact inspection, as-of projection, period interpretation, metric resolution, optional derivation/verification, and export. Raw responses remain distinct from interpretations. URL, response hash, JSON pointer, and accession connect a decision to preserved evidence.

Fiscal inference uses admitted annual windows and cumulative boundaries; raw fiscal-focus labels need not label comparative facts correctly. Classifying one annual context differs from resolving a fiscal-label query across candidates. Other uncertain contexts can block selection, a usability cost illustrated by H&R Block.

Instant metrics require caller-supplied period-end associations. An exact date does not override eligibility, and the caller remains responsible for its fiscal-label interpretation. Reproducibility cannot certify an erroneous human association.

4.2 Reported metrics and conservative arithmetic

Explicit concept mappings support revenue, net income, assets, liabilities, cash and cash equivalents, and operating cash flow. Competing supported scopes can conflict; plausible values do not make concepts interchangeable. Missing liabilities are not silently synthesized as assets minus equity.

Reported and derived values have different evidentiary status. A reported scalar is selected from eligible observations under the metric contract. A derived quarter is an arithmetic reconstruction from reported cumulative operands. The permitted formulas are Q2 as six-month cumulative minus Q1, Q3 as nine-month cumulative minus six-month cumulative, and Q4 as annual minus nine-month cumulative. The derivable metrics are revenue, net income, and operating cash flow. Instant values and Q1 are not reconstructed through these formulas.

Subtraction requires matching issuer, metric, exact concept, unit, fiscal start, compatible intervals/calendar, cutoff, policy, and snapshot. Both operands must resolve canonically. Exact arithmetic preserves source numbers without increasing measurement precision or proving identical original dimensional context.

Individually selected latest values can have incompatible histories. Changed latest operands without shared-filing corroboration can return `unpaired_value_revision`; unpaired amendments also conflict. Ordinary cross-filing subtraction is not categorically forbidden. Shared filing is a conservative guard, not proof that every accepted pair is comparable or every rejected pair wrong.

Derived readiness includes supporting calendar evidence as well as operands. It is a bound for the selected evidence, not the earliest possible public derivation. Reported-then-derived selection never uses arithmetic to bypass reported conflict or ambiguity.

4.3 Explicit states and evidence inspection

Each requested panel cell persists as a row. Resolved reported and resolved derived states expose scalar values. Other states distinguish unavailable evidence, period ambiguity, insufficient evidence, conflict, unsupported requests, and source-verification failure. These are computational decisions under a contract, not diagnoses that the issuer's accounting is wrong. A conflict may arise from competing concepts or revisions; unavailable may mean a filing had not yet become eligible.

States partition the requested cells, whereas reason categories can overlap. A row may have more than one reason for remaining unresolved. Analysts must not sum reason counts as if they represented distinct companies or observations. Keeping unresolved rows also preserves the requested denominator. If users later filter to resolved rows, that exclusion becomes their explicit research decision rather than an invisible effect of data ingestion.

Inspection retains considered, selected, and rejected observations, accessions, temporal decisions, calendar support, revision groups, and operands. This enables challenges to values and refusals; provenance itself does not certify correctness.

4.4 Immutable replay and portable outputs

Pinned execution selects one exact snapshot and uses an offline, read-only evidence path. A newly captured snapshot does not alter an already pinned request. Receipts bind query policies, represented rows, evidence identities, and software contracts. Replay checks fail explicitly when required objects are missing or corrupt, the receipt is tampered with, software is incompatible, or reproduced semantics differ. Unresolved decisions are part of the reproduced object.

The relevant notion of identity is semantic rather than universal byte identity. Execution timestamps and performance measurements are not the financial meaning of a panel. CSV, Parquet, and DuckDB exports preserve exact value text with explicit numeric kind and provenance metadata. Their container bytes need not be identical for represented rows to agree. Conversely, equal visible numbers do not establish equal evidence decisions if revision policies or supporting sources differ.

The software fingerprint covers Python source, Python version, httpx, and simplejson. It does not exhaustively pin the OS, timezone database, or export dependencies. Long-term replay still requires accessible evidence and compatible software.

5. Study design and audit procedure

The empirical design is a repository-centered artifact audit with historical benchmark recounting and current regression testing. A supplementary audit checks archived outputs without rebuilding the large historical panel and records the inspected commit and current versus historical software identities. The local regression run performed during paper preparation passed all 814 tests in 284.79 seconds; its log is preserved separately from historical reports. These distinctions make the provenance of the study's evidence explicit.

The scale population consists of 100 included ordinary US-GAAP operating companies grouped into 14 manually assigned categories. The universe manifest additionally lists excluded records, so counting every manifest row would overstate the experimental population. Banks, insurers, broker-dealers, funds, and IFRS issuers are outside the supported benchmark. Selection is purposive. It is not a probability sample, historical security master, or representative sample of the US market. Current ticker and name metadata do not establish historical tradable security identity.

The requested grid crosses six metrics, fiscal years 2023 and 2024, five periods - FY and Q1 through Q4 - two UTC cutoffs, and two revision policies. The cutoffs are 2024-08-15T00:00:00Z and 2025-04-01T00:00:00Z; policies are first-reported and latest-available. Multiplication gives $100 \times 6 \times 2 \times 5 \times 2 \times 2 = 24{,}000$ cells. Policy and cutoff variants are separate requests, not independent economic observations. In particular, overlapping annual and quarterly requests, and repeated instant dates, create dependence within the grid.

Two disjoint archived jobs each cover 50 issuers. Their selected snapshot is `7ee0bbe49b267ce50d63bf1f45bc4f62b017a195db7e2fcefe99c3dc68669ef0`. The local corpus contains 406 raw artifacts totaling 530,856,505 bytes. Its manifest declares partial evidence because aggregate acquisition did not include original XBRL; object integrity must not be confused with source completeness. The supplementary audit checks each object's saved size and hash, verifies archived output-file hashes, checks ledger identities, reconstructs state and subgroup counts from saved rows, and confirms the combined semantic dataset hash. These procedures establish consistency with preserved artifacts; they do not independently reconstruct the financial truth of all observations.

Validation combines literal golden expectations, instrumented invariants, future-evidence mutations, replay, and original-XML verification. Current tests exercise fixtures and synthetic cases. These layers answer different questions and are not interchangeable.

The golden inventory contains 255 authored expectations: 32 canonical, 25 authentic derived-quarter, 156 broad, 22 calendar, and 20 revision contracts. Policy/cutoff variants count separately. Source-authored expectations provide within-project separation from resolver answers, not independent investigators or unbiased case selection. Additional operand, source, and replay tests are excluded from this inventory.

Recovery found four Git commits with 63 Python files matching the historical fingerprint and a matching a1 wheel. Installation with saved dependencies in Python 3.12.14 matched the recorded software identity. One new offline replay recomputed 120 rows with its original

receipt. Multiple matching commits prevent identification of a unique original whole checkout or machine. The 200-replay total remains historical; supporting materials also document a separately executed six-cell current example.

Figure extraction verifies shard and combined semantic row hashes before counting states and metric/period groups. Policy pairs join on CIK, metric, fiscal year, period, and cutoff; uniqueness and equal key sets are checked. Exact numerical changes are directly counted, not inferred from aggregate coverage. Companion CSVs preserve denominators, source hashes, and rendering requirements.

No randomized comparison, causal estimate, market-wide error rate, or fresh full benchmark is reported. Baselines and independent adjudication remain proposed extensions.

AI assistance. OpenAI Codex assisted with repository inspection, literature discovery, evidence extraction, drafting, editing, typesetting, and preparation of review, website, and submission materials. The figures visualize verified archived data. The human author remains responsible for checking sources, understanding the analysis, and approving claims and the final manuscript. Human contributions to the underlying software and research, and any other AI use, remain to be declared; no independent human validation is asserted.

6. Results and their interpretation

The archived row recount confirms 16,706 resolved cells out of 24,000, or 69.61% scalar coverage after rounding. Reported values account for 14,136 cells and derived values for 2,570. The remaining 7,294 cells preserve explicit conservative outcomes. Figure 1 gives the complete state partition, including the zero source-verification-failed count. That zero reflects the panel configuration and must not be read as universal original-source verification.

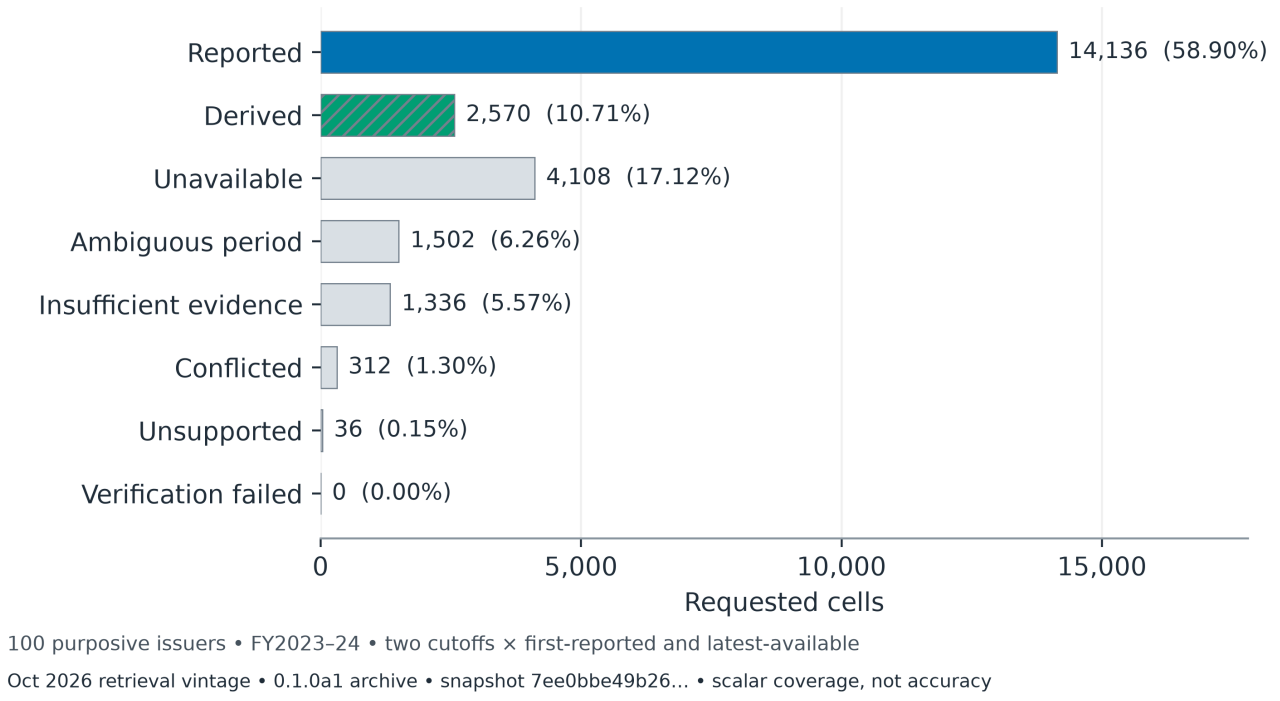


Figure 1. Complete result-state composition. Denominator: 24,000 requests across the 100 purposively selected issuers, six metrics, FY2023-2024, five periods, both cutoffs and both revision policies. Counts and percentages retain unresolved requests.

Figure scope. Figures 1-4 describe the archived 0.1.0a1 experiment using snapshot 7ee0bbe49b267ce50d63bf1f45bc4f62b017a195db7e2fcefe99c3dc68669ef0, retrieved during 7-8 October 2026 UTC. Historical cutoffs are 15 August 2024 and 1 April 2025, both 00:00 UTC. First-reported and latest-available requests are pooled except where explicitly matched. Repeated requests are dependent; these figures have no sampling confidence intervals and estimate neither accounting accuracy nor market-wide behavior.

Coverage varies by metric. The assets metric resolves most frequently, while liabilities resolves least frequently in this grid. Revenue contributes 286 of the 312 conflicts, and liabilities has 1,900 unavailable cells. These observations fit the implementation's exact-concept and competing-scope restrictions, but they are descriptive evidence rather than a causal decomposition. An absent scalar can reflect cutoff timing, unsupported concepts, or additional interpretive constraints.

2 | Reported and derived contributions differ by metric

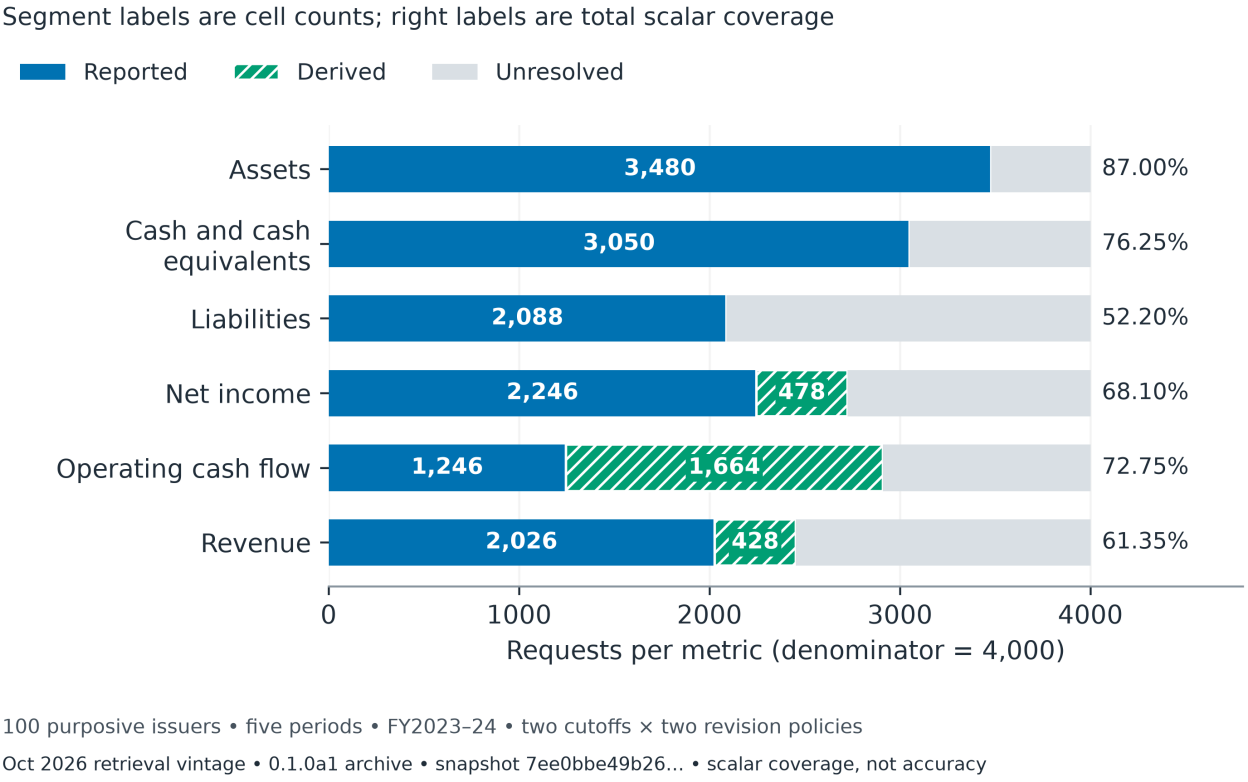


Figure 2. Metric-specific scalar coverage and its components. Each metric has 4,000 requests in the Figure 1 population. Solid and hatched segments distinguish reported and derived counts; right labels give total scalar coverage. Evidence vintage and policy scope follow the shared note above.

Operating cash flow has 1,664 derived resolutions compared with 1,246 reported resolutions, whereas the three instant metrics have no derived component. This composition matters for interpreting the same headline coverage: different metrics depend on different kinds of evidence. The chart is a decomposition of the existing reported-then-derived output, not an executed ablation showing how another resolver would perform.

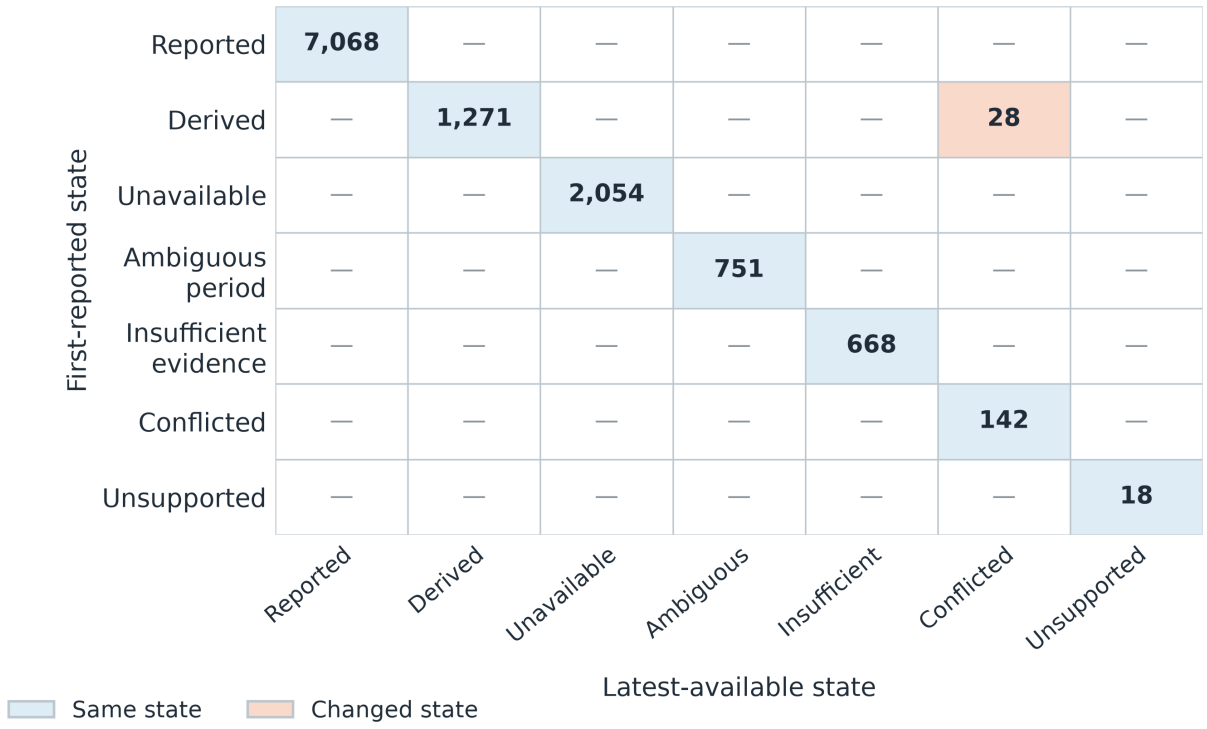
First-reported requests resolve 8,367 of 12,000 cells; latest-available requests resolve 8,339. Directly matching the rows shows that all 28 state changes move from derived to conflicted; no pair gains a scalar. The remaining 11,972 pairs keep the same state. However, state stability does not establish value stability: among 8,339 pairs resolved under both policies, 146 have different numerical values, or 1.75%. Exact decimal comparisons retain units and receipt identifiers. These are repeated request pairs, not 146 independent restatement events. Nor does this comparison establish which policy is economically preferable.

This matched comparison establishes where state changes occur, but does not isolate the causal effect of the revision-pairing guard. The archive also records 23 incompatible-revision-operand findings; that different diagnostic denominator must not be substituted for the 28 matched request transitions. A guard-specific ablation would be needed to quantify its separate effect. No downstream regression or portfolio effect is measured.

3 | Matching policies reveals 28 lost scalar results

12,000 matched requests; 28 move from derived to conflicted; none gain a scalar

Among 8,339 pairs resolved under both policies, 146 change numerical value (1.75%)



100 purposive issuers • match: CIK, metric, FY, period, cutoff • dash = zero pairs

Oct 2026 retrieval vintage • 0.1.0a1 archive • snapshot 7ee0bbe49b26… • scalar coverage, not accuracy

Figure 3. Matched policy transitions. Rows show first-reported states and columns latest-available states for 12,000 pairs matched by CIK, metric, fiscal year, period and cutoff. Dashes denote zero. The all-zero source-verification-failed row and column are omitted visually but retained in the CSV. Numerical-change percentage uses the separate denominator of 8,339 jointly resolved pairs. Evidence vintage follows the shared note above.

Period-level grouping reveals variation concealed by metric totals. Q4 net income resolves 526/800 requests (65.75%), including 478 derived values. Q4 operating cash flow resolves 580/800 (72.50%), all derived. Equal FY and Q4 counts for instant metrics reflect requests at the same mapped year-end dates in this grid, not independent confirmation from two accounting observations. The heatmap describes availability under fixed cutoffs and contracts; it does not diagnose why every missing cell is unresolved or rank quarterly accounting quality.

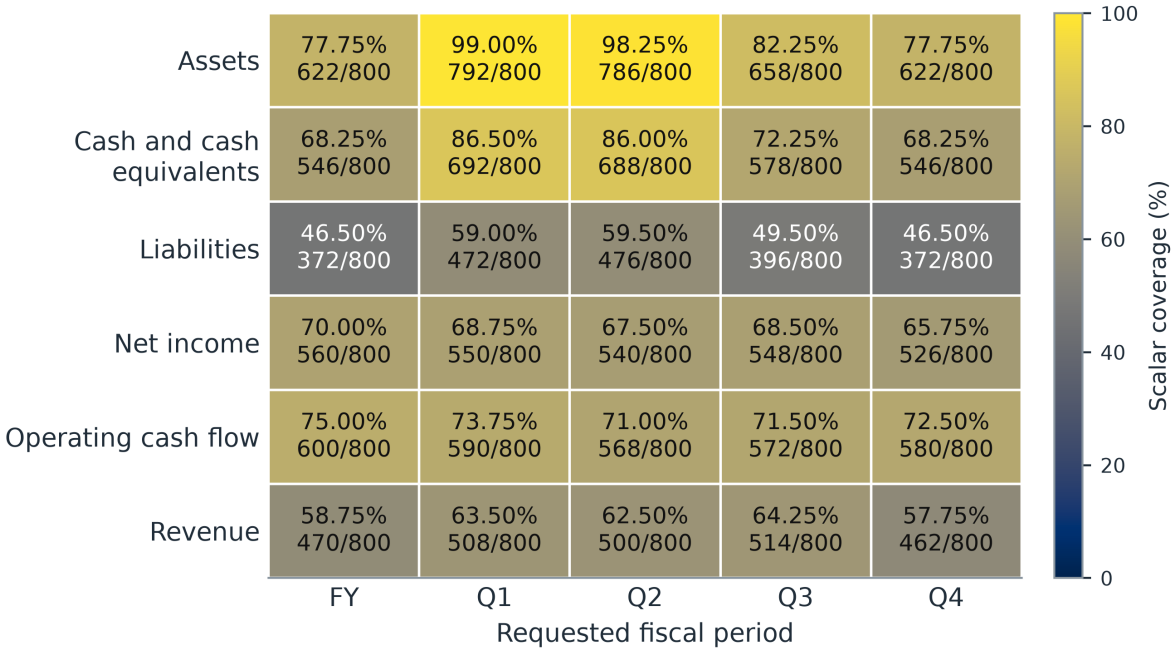
The archived reports record 2,435,238 invariant checks, 200 successful panel receipt replays, and zero unexpected mismatches. These are counts of instrumented checks and replay operations, not independent correctness trials. Repeated assertions over related cells can be numerous without materially increasing independent coverage of accounting situations.

Similarly, the golden inventory supplies focused contract evidence, not a random audit of every resolved scalar.

The temporal audit records 500 non-vacuous comparisons across all 100 issuers. Five comparisons per issuer concern earlier records, calendar interpretation, and three revision-policy views. Ninety-eight issuers use future net-income observations; two use operating cash flow where future net-income observations were absent. The saved audit verifies that mutation populations were nonempty. This is stronger than counting an empty mutation as a pass, but remains evidence about the specified transformations rather than a proof of universal freedom from look-ahead bias.

4 | Availability varies jointly by metric and period

Each tile has 800 requests; exact counts accompany percentages



100 purposive issuers • FY2023–24 • two cutoffs × first-reported and latest-available

Oct 2026 retrieval vintage • 0.1.0a1 archive • snapshot 7ee0bbe49b26… • scalar coverage, not accuracy

Figure 4. Metric-by-period scalar coverage. Every tile has 800 requests: 100 issuers × two fiscal years × two cutoffs × two policies. Labels give percentages and exact resolved counts. The fixed color scale spans 0-100%; numbers make interpretation independent of color. Evidence vintage follows the shared note above.

Original-source evidence is substantially narrower. Of 200 separate capture-vintage verification queries, six are uniquely verified and seven have multiple equivalent matches. The remaining 187 report unavailable instances. Accordingly, verified coverage is 13/200, or 6.5%, for that query set. It is neither 6.5% nor 100% accuracy for the financial panel. The source audit and historical financial grid have different denominators and temporal contracts.

Operational records report job wall times of 2,353.23 and 2,328.24 seconds, with per-process peak memory of 880.03 and 976.11 MiB. The jobs ran concurrently with other validation work, so these figures are contextual resource observations. They establish neither isolated throughput nor a speedup. Completed-job resume records report verification and skipping of all 100 issuers with unchanged semantic files; interruption equivalence is tested separately. Retaining cumulative rows and provenance also means checkpoint storage and memory can grow with the requested grid.

7. Calendar and revision case studies

7.1 H&R Block's shortened transition

H&R Block's filing identifies a transition from May 1 to June 30, 2021. Inclusive date arithmetic gives 61 days. The frozen calendar ledger tests that this context remains an `other_duration` with no manufactured fiscal-year or quarter label, while surrounding full annual contexts retain supported classifications. Before the relevant filing becomes eligible, the transition observation remains excluded. The source narrative supports identifying a transition; it does not authorize forcing it into an ordinary quarter. [10]

This case exposes the risk of using current fiscal-year-end metadata as historical truth. A June year end visible in later submissions cannot, by itself, relabel an earlier April-based year. FinPanel keeps that comparison as a retrospective annotation outside the admitted calendar. Mutation tests change later observations and metadata and require earlier admitted interpretations to stay stable. The claim is about that implemented boundary, not a general solution to every fiscal transition.

The case also demonstrates conservative under-resolution. Correctly classifying some annual contexts does not guarantee a scalar for every fiscal-label query. Other uncertain comparative or quarterly contexts can still block selection across candidates. This distinction matters because it identifies a specific source of missingness that higher-level coverage tables would otherwise conceal. No authentic extended transition-year case was established in the audited evidence.

7.2 Week-based calendars and leap years

The calendar ledger includes 371-day annual contexts for Costco and Target, a 364-day Target annual context, a 98-day Target final quarter, and a 366-day Nike annual context. These literal date intervals illustrate why universal assumptions of 365-day years or 90-day quarters are inadequate. They do not imply that duration alone supplies fiscal identity. Context, filing association, and admitted boundary evidence remain necessary.

The 98-day quarter is particularly instructive: a longer final quarter can be compatible with a week-based year, while a 61-day transition needs a different interpretation. A rule that merely chooses the nearest conventional quarter length could conflate the two. The repository's cases instead preserve the observed interval and test classification together with filing-cutoff eligibility. Evidence about these particular calendars should not be generalized to all issuers using weekly accounting calendars.

7.3 Mattel's revised 2017 evidence

Mattel's amendment, accession `0001628280-19-013975`, contains narrative support for a specifically identified restatement of 2017 quarterly information and separately discusses revisions to other periods. This permits a narrow formal-restatement attribution. A changed aggregate observation alone would not establish its accounting cause. The paper therefore uses the issuer narrative for that case and avoids labeling every change in the repository a restatement. [11]

The quarter-derivation test supplies an interpretable numerical contrast. Original annual net income of −1,053,836,000 USD and nine-month net income of −772,553,000 USD imply a Q4 residual of −281,283,000 USD. The amended pair, −1,054,579,000 and −879,286,000 USD, implies −175,293,000 USD. Both amended operands occur in the same filing. The arithmetic differs because the evidence differs; the result remains a derived residual even where reported quarterly evidence offers a separate comparison.

At noon UTC on November 11, 2019, the test's latest-available result retains the original residual. At noon UTC on November 13, the revised pair is eligible under latest-available, while first-reported retains the original pair. All-available exposes incompatible versions and remains

conflicted. These outcomes illustrate that a revision policy is part of the research question, not a cosmetic output option. Neither original nor revised values are universally preferable for every empirical design.

7.4 Incompatible revision operands

Mattel's FY2018 example supplies a complementary refusal. Original annual and nine-month amounts are −530,993,000 and −545,906,000 USD, yielding 14,913,000 USD. Revised amounts are −533,299,000 and −542,938,000 USD, but their accessions differ: the annual revision is in 0001628280-19-013975, while the cumulative revision is in 0001628280-19-013987. Subtracting them would produce 9,639,000 USD. FinPanel withholds that plausible result under latest-available because the selected revised operands lack required pairing evidence.

This is the clearest example of arithmetic sufficiency differing from evidentiary sufficiency. The program knows how to subtract the numbers; the contract refuses to assert a coherent quarterly reconstruction. First-reported preserves the original residual. Whether an expert could justify the revised pair using additional narrative is an open accounting question, not something the program has disproved. Related Kraft Heinz cases retain changed comparative observations while declining to infer a cause automatically.

8. Discussion and threats to validity

RQ1 is answered at the implementation level: observations are bounded before calendar and revision inference, with separate retrieval identity. Tests cover specified timestamp, date-only, metadata, and mutation boundaries. Acceptance remains a proxy, and modern aggregates cannot establish historical delivery.

RQ2 is answered descriptively: 69.61% scalar coverage and structured missingness under declared contracts. Coverage is not accuracy. Accuracy requires an independently defensible reference for decisions, including justified abstentions. Unsupported guesses can raise coverage; unnecessary refusals can lower it. Neither establishes better data.

Internal validation can share project misconceptions. Frozen ledgers are not independent adjudication; sparse XML verification cannot settle all accounting interpretations. Human review should challenge resolved and unresolved decisions and distinguish source, timing, scope, and interpretation disagreements.

RQ3 is conditional: receipts can repeat evidence-bearing decisions, including unresolved ones, but cannot establish truth or eliminate omitted sources. The recovered implementation and one replay do not demonstrate independent replication, a unique machine identity, or exhaustive dependency preservation. Public summaries and hashes cannot reconstruct absent raw objects.

Purposive selection, restricted metrics, and no historical security universe limit external validity. Category differences cannot support causal industry explanations. Financial institutions, IFRS issuers, segment aggregation, currency conversion, and returns are outside scope. No baseline or commercial-provider comparison supports superiority or quantified bias reduction.

Optional extensions include fixed-evidence baseline comparisons, independent stratified adjudication of scalars and refusals, and investigator-owned replay. Calendar and revision-guard ablations could isolate their effects; reported-only comparisons could assess incremental availability. These experiments were not run and are not prerequisites for the present descriptive claims.

Investment applications would additionally require historical security membership, prices, execution assumptions, and delivery evidence. No portfolio benefit is inferred.

9. Conclusion

FinPanel demonstrates a bounded approach to constructing SEC fundamentals in which temporal eligibility, fiscal interpretation, revision policy, and derivation evidence remain visible. The archived population yields 16,706 resolved cells and 7,294 explicit unresolved outcomes. Calendar transitions and incompatible revised operands show why a defensible pipeline sometimes needs to withhold a scalar. Immutable snapshots and receipts support repetition of those decisions while preserving their policies and evidence identities.

The supported conclusion is an infrastructure result: evidence-preserving decisions can be implemented, inspected, and replayed under explicit contracts. It is not an estimate of market-wide accuracy, a reconstruction of historical vendor delivery, or a demonstration of investment benefit. Independent assessment is needed to evaluate the accounting adequacy of the featured decisions and the limits of the evidence. Comparative baselines are needed for superiority or quantified bias-reduction claims, which this paper does not make. Full historical evidence review requires authorized access to the local corpus.

Availability and declarations

Data and code availability (archive identifiers pending). A local deposit candidate accompanies this draft: processed records for all 24,000 requests, figure data and scripts, archived validation reports, source manifests, frozen case evidence, and versioned software. It supports recalculation of descriptive summaries and figure regeneration without the large SEC corpus. Full historical financial replay still requires the excluded 406-object snapshot; source manifests cannot replace its bytes. Repository DOI, data/code licences beyond the existing software licence, and the authorised raw-evidence access arrangement remain unresolved. Nothing has been deposited publicly by this preparation.

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Declarations

Author contributions. Finley Dou is the sole author. He designed the study, wrote the FinPanel software it audits, carried out the analysis, prepared the figures and wrote the manuscript, and approves this version. CRediT: Conceptualization, Software, Methodology, Investigation, Data curation, Formal analysis, Visualization, Writing - original draft, Writing - review and editing.

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Competing interests. The author wrote FinPanel and maintains it. This paper evaluates that software, so the author is not a disinterested assessor of it, and the evidence is presented so that a reader can check the reported behaviour independently rather than take the assessment on trust. The author also holds a role in PRISM's Reviewer Corps and is a co-author on a separate article published by this journal; he took no part in refereeing or deciding on this manuscript. The referee, Henry Wang, is Editor-in-Chief of PRISM and is a co-author of the

author on a separate article published by this journal in the same month. That relationship is disclosed here rather than avoided, and readers should weigh the review accordingly.

Ethics. Not applicable. The work analyses public SEC filings and software behaviour. No human participants, animal subjects, personal data or laboratory procedures are involved.

Use of artificial intelligence. OpenAI Codex assisted with repository inspection, literature discovery, evidence extraction, drafting, editing, typesetting and the supporting materials. The author wrote the FinPanel software itself, designed the study, and verified the analysis and the reported numbers. Codex is not an author.

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Data and code availability. The processed data, analysis code and verification materials behind every figure and statistic in this paper are deposited with the journal at https://prism-journal.github.io/data/finpanel-deposit.zip (18 MB). This is a journal-hosted archive rather than a repository with a persistent identifier, so it carries no DOI. The 406-object raw snapshot (530,856,505 bytes) is excluded from that deposit in any case, so the originating experiment cannot be re-run in full from it; deposit/claim-data-crosswalk.md records what each reported claim rests on. Readers can therefore recheck the reported numbers against the deposited records.

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